

QTSR/RW – 24 / 1667

BIOLOGY FOR ENGINEERS

[Max. Marks : 60]

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated against them.
- (3) Due credit will be given to neatness and adequate dimensions.
- (4) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1.
 - (a) Illustrated fundamental differences between science and engineering through case studies in the context of biological sciences ? 5(CO1)
 - (b) Explain the hierarchy of the levels of organization from cells to organisms. Provide examples to illustrate each level. 5(CO1)
2.
 - (a) Describe the structure and function of carbohydrates, proteins, and lipids as essential biomolecules. How do these molecules contribute to the basic building blocks of life ? 5(CO2)
 - (b) Discuss the classification of enzymes and their significance in biochemical processes within living organisms. 5(CO2)
3.
 - (a) Explain the concept of entropy changes and free energy in living organisms. 5(CO3)
 - (b) Describe the process of synthesis and breakdown of glucose in terms of metabolism and bioenergetics. 5(CO2,3)
4.
 - (a) Concept of genetic codes and the expression and transmission of genetic information. How do these processes contribute to the diversity of life? 5(CO4)

Contd.

- (b) Explain the concept of DNA cloning and discuss its applications in modern biotechnology. 5(CO4)
- 5. (a) Discuss the differences between resting and action potentials and their significance in biological systems. 5(CO5)
- (b) Describe the propagation of bio-electric signals and provide example utilized in medical diagnostics ? 5(CO5)
- 6. (a) Explain the principles of bio-sensors. How are bio-sensors used in various applications ? 5(CO5)
- (b) Discuss the principles behind electro physiological signal analysis and its role in medical diagnostics in real world scenario. 5(CO5)

